

What Factors Drive a Movie's Financial Success?

A correlational analysis of budget, genre, and audience reception on a Movie's profitability

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1 Introduction

Statement and Objective

Billions of dollars are spent on film production every year. Additionally, when a movie has significant cultural impact, it becomes a central topic of public discussion, and even related movies and YouTube videos receive millions of views. In addition, as there are various jobs related to film production, many people are in the industry, so it has a critical financial and professional impact on numerous stakeholders. Therefore, We believe that film companies need to analyze what the most important factor in film production in order to achieve success with limited resources.

To address this need, our projects research question is to investigate which critical factors (such as budget, public interest, audience ratings, or genre) has the most substantial impact on a movie's financial outcome. Rather than simply evaluating gross revenue, we focus on net profit (Revenue - Budget) and the ratio between net profit and budget as the core target metric to better reflect true commercial success. Using the TMDB 5000 Movie Dataset, we perform exploratory data analysis and statistical modeling to provide data-driven insights into how film production studios can optimize their resources for maximum profitability.

2 Data

Data Sources

The core dataset used for this report is the TMDB 5000 Movie Dataset which we found off of Kaggle.com posted by The Movie Database (TMDB) and one other collaborator 9 years ago in 2017. The data set contains 4803 rows each accounting for an individual movie entry, and 20 columns that specify notable details of the movies. The following are the names of each column:

- "budget", "genres", "homepage", "id", "keywords", "original_language", "original_title", "overview", "popularity", "production_companies", "production_countries", "release_date", "revenue", "runtime", "spoken_languages", "status", "tagline", "title", "vote_average", "vote_count".

In this analysis however we won't be using all of these columns and will primarily focus on using budget, revenue, popularity, vote_average, vote_count, genres, production companies, release_date, title, id.

[1]	"budget"	"genres"	"homepage"
[4]	"id"	"keywords"	"original_language"
[7]	"original_title"	"overview"	"popularity"
[10]	"production_companies"	"production_countries"	"release_date"
[13]	"revenue"	"runtime"	"spoken_languages"
[16]	"status"	"tagline"	"title"
[19]	"vote_average"	"vote_count"	

Note: The columns were stored as JSON-formatted strings instead of plain text, so they need to be parsed before they can be used. So we use json functions to splice the strings.

Analysis of Data Cleaning:

Among the 4,803 movies in the dataset, 1,574 movies (32.77%) were dropped due to missing or zero values in either their reported budget or revenue. Based on the visualizations and summary tables, this data loss is not entirely random. The dropped observations tend to be concentrated in older release periods (particularly pre-2000s) where financial tracking was less publicly disclosed. Furthermore, the missing data is heavily prevalent among relatively minor categories such as Documentaries and Foreign films, as well as lower-budget independent productions. In contrast, modern mainstream blockbusters tend to have complete financial records.

Exploratory Data Analysis

Total initial records: 4,803

Movies dropped (missing/zero budget or revenue):
1,574 (32.77%)

Remaining movies after data cleaning: 3,229

Figure 1: Distribution of Retained vs. Dropped Movies by Release Year

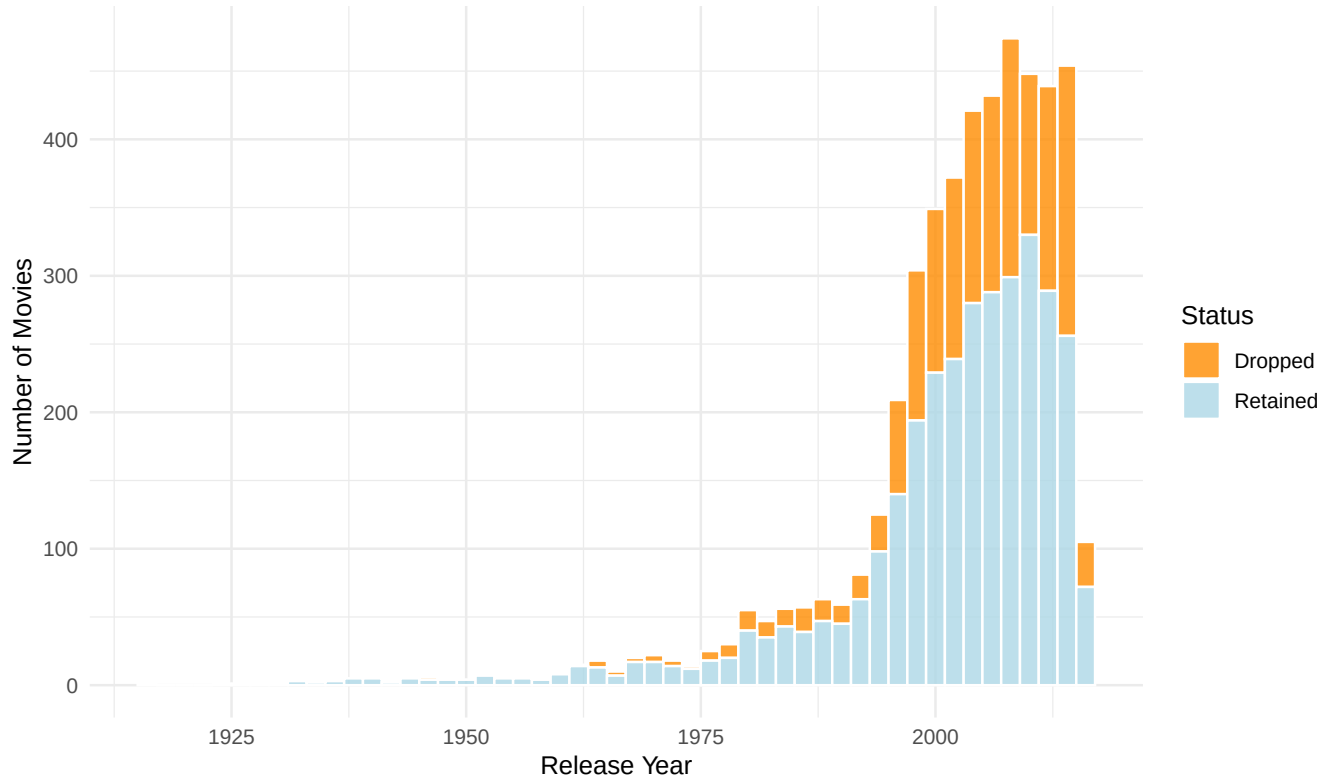
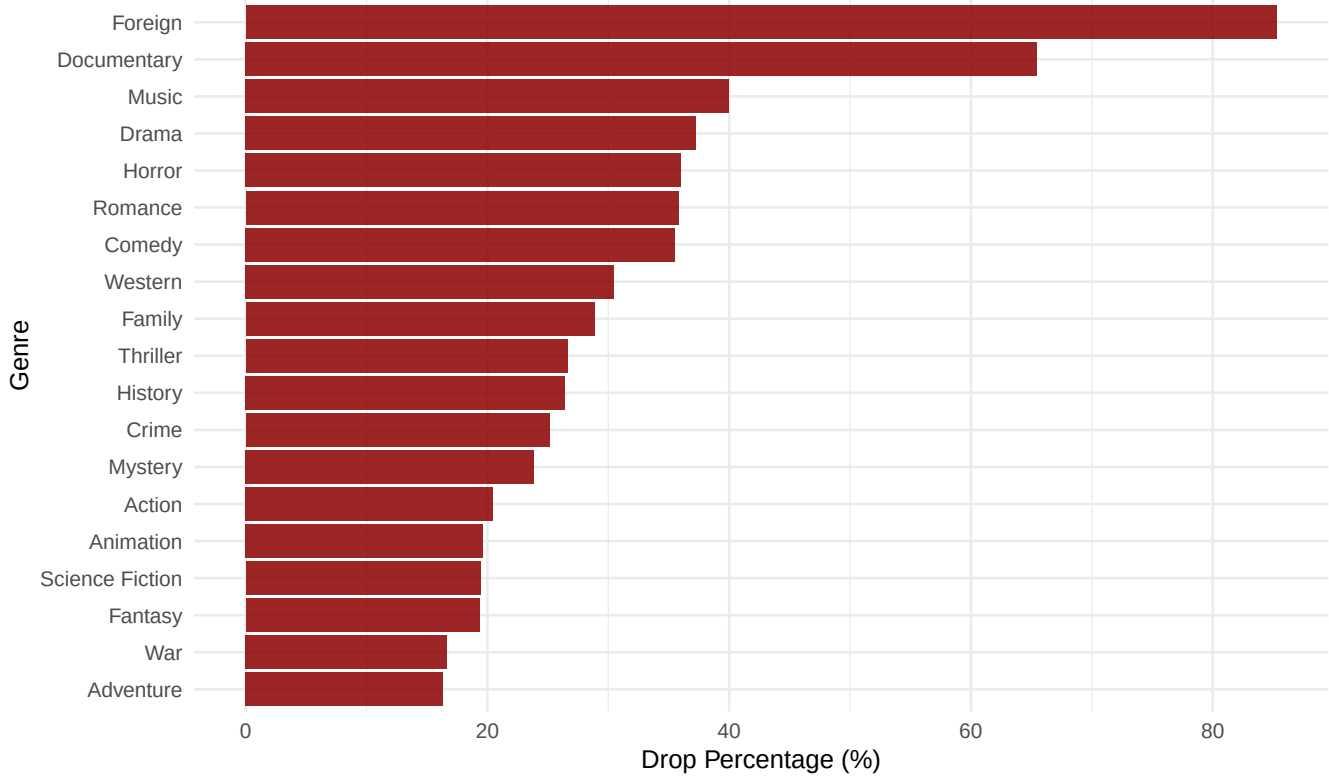


Table 1: Top 10 Genres by Dropped Data Percentage

Genre	Total Movies	Dropped	Retained	Drop Rate (%)
TV Movie	8	8	0	100.0
Foreign	34	29	5	85.3
Documentary	110	72	38	65.5
Music	185	74	111	40.0
Drama	2297	856	1441	37.3
Horror	519	187	332	36.0
Romance	894	320	574	35.8
Comedy	1722	612	1110	35.5
Western	82	25	57	30.5
Family	513	148	365	28.8

Figure 2: Data Drop Rate by Genre (Budget/Revenue Missing or 0)



Analysis of Exploratory Data:

After isolating the clean and usable data, the distribution of movie profits exhibits a strong right skew as seen in figure 1. The vast majority of films cluster around modest profit margins, while a select few blockbuster outliers stretch the tail with exceptionally high returns. When analyzing genre representation, “Drama” and “Comedy” are the most frequently occurring genres, providing a robust sample size for statistical analysis as seen in table 1. On the other hand, niche genres like “TV Movie,” “Foreign,” and “Documentary” are less represented. Because these underrepresented genres have too few observations to support reliable comparisons, any conclusions drawn regarding their average profitability or financial trends will be interpreted with extreme caution.

3 Methodology

Net Profit (calculated for each movie):

$$\text{Profit} = \text{Revenue} - \text{Budget}$$

Return on Investment (calculated for each movie):

$$\text{ROI} = \text{Profit} / \text{Budget}$$

Pearson correlation coefficient, computed pairwise between budget, popularity, vote_average, vote_count, and profit, where x and y are each pair of variables being compared, $\bar{x}$ and $\bar{y}$ are their means, and x_i , y_i are individual movie observations:

$$r = \text{sum}((x_i - x_{\text{bar}}) * (y_i - y_{\text{bar}})) / \text{sqrt}(\text{sum}((x_i - x_{\text{bar}})^2) * \text{sum}((y_i - y_{\text{bar}})^2))$$

Ordinary least squares (OLS) regression line, fit between vote_average and profit, where y_hat is predicted net profit, x is audience rating (vote_average), B0 is the intercept, and B1 is the slope:

$$y_{\text{hat}} = B0 + B1 * x$$

Genre-level summaries (mean/median profit, mean popularity) are computed on a long-format table with one row per movie-genre pair, since movies can belong to multiple genres.

4 Results

4.1 Part I: Which Factors Correlate Most With Profit?

Figure 3: Correlation Between Movie Factors and Outcomes

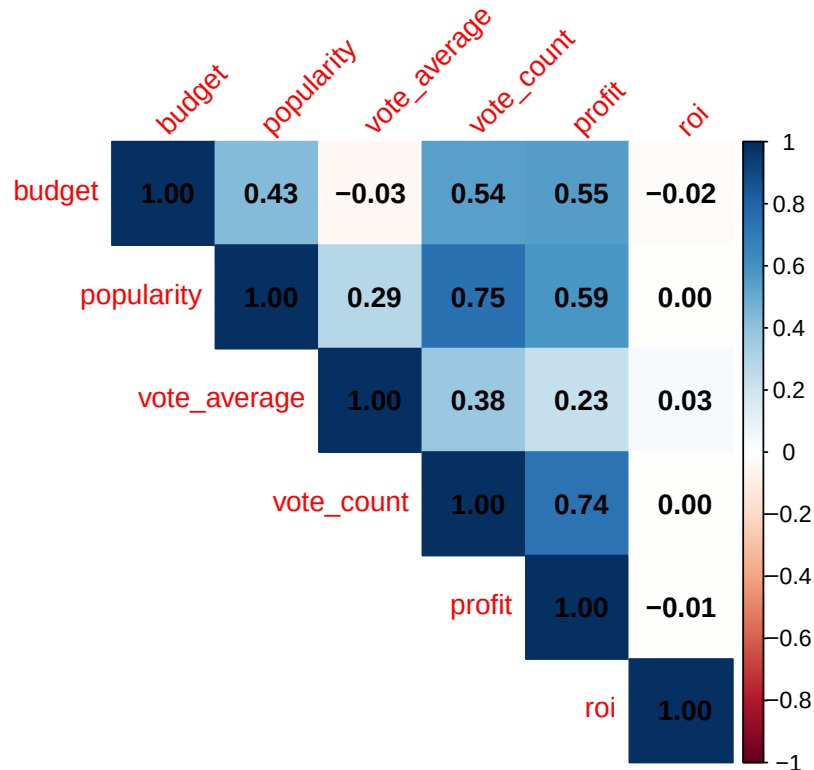


Table 2: Pairwise Correlations Between Movie Factors and Outcomes

Variable	budget	popularity	vote_average	vote_count	profit	roi
budget	1.00	0.43	-0.03	0.54	0.55	-0.02
popularity	0.43	1.00	0.29	0.75	0.59	0.00
vote_average	-0.03	0.29	1.00	0.38	0.23	0.03
vote_count	0.54	0.75	0.38	1.00	0.74	0.00
profit	0.55	0.59	0.23	0.74	1.00	-0.01

Variable	budget	popularity	vote_average	vote_count	profit	roi
roi	-0.02	0.00	0.03	0.00	-0.01	1.00

Analysis of Correlation Matrix:

When analyzing the relationship of the factors, `vote_count` shows the strongest correlation at $r = 0.74$, followed by `popularity` with a $r = 0.59$ and `budget` with a $r = 0.55$. Notably `vote_average` (audience rating) is weakly correlated with `profit` at $r = 0.23$. This suggests that how much a film is able to generate hype/attention for a film, as shown through vote count and popularity tracks financial success more closely than how well it is actually rated by viewers. Meaning being seen and talked about is more tied to profitability than a movie being rated good or bad by viewers. However something important to note is that profit and ROI tell very different stories, as none of the factors show a meaningful correlation with ROI with all of them having a $r < 0.04$, including budget itself with a $r = -0.02$. Meaning that while bigger-budget and more popular films tend to generate more profit when strictly talking about a movies revenue, they are not more efficient at doing so relative to budget of the film. So larger investments do not reliably give proportionally large returns. This is heavily important in the matters of interpreting “success” as a factor that predicts high profit does not necessarily predict a good return on investment.

Figure 4: Average Net Profit by Genre

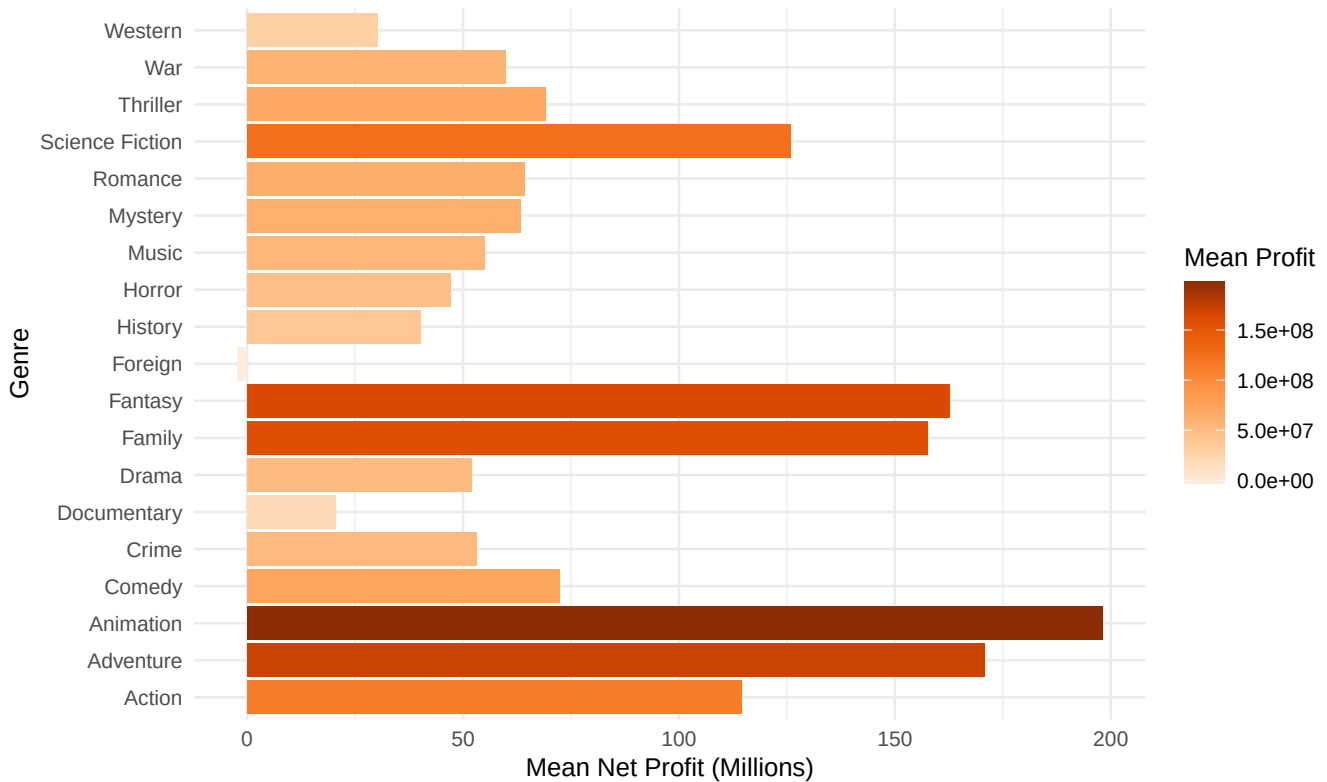


Table 3: Genre-Level Summary: Movie Count, Profit, and Popularity

	Genre	N Movies	Mean Profit	Median Profit	Mean Popularity
8	Animation	188	198090651	125484870	46
2	Adventure	661	170714153	65283742	45

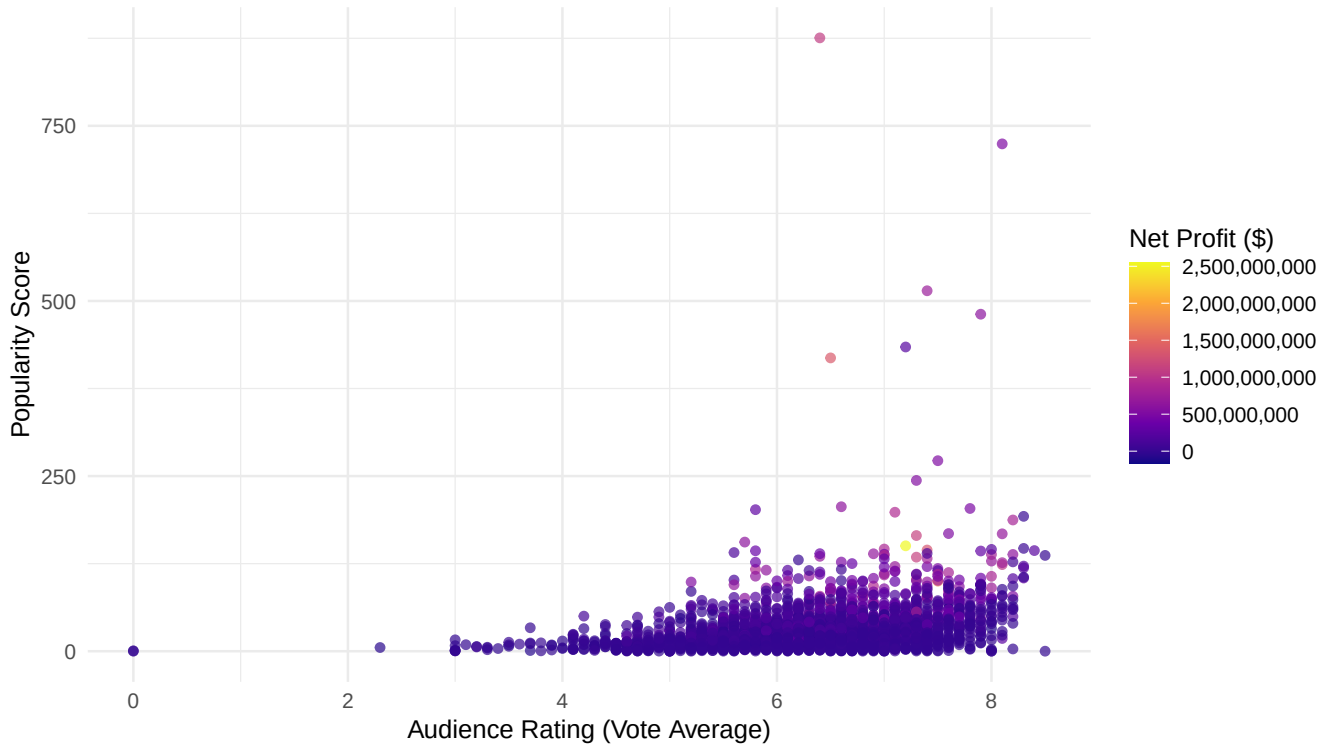
	Genre	N Movies	Mean Profit	Median Profit	Mean Popularity
3	Fantasy	342	162675510	62653067	43
9	Family	365	157588117	65918663	36
4	Science Fiction	431	125956783	35489265	43
1	Action	918	114424359	45189732	37
11	Comedy	1110	72323910	26229098	25
7	Thriller	935	69233753	25944725	31
12	Romance	574	64263807	19643680	21
14	Mystery	265	63308118	25283321	30
16	War	120	59794160	19955685	28
17	Music	111	55094012	20550513	18
5	Crime	521	53147816	20965854	28
6	Drama	1441	51943285	14541707	25
13	Horror	332	47169591	25545686	25
15	History	145	40162408	20500000	22
10	Western	57	30290807	5795189	23
18	Documentary	38	20408260	6208945	6
19	Foreign	5	-2249849	-376209	0

Analysis of Genre vs. Profit:

Animation shows both the highest mean at \$198.1M and median at \$125.5M profit, with the two staying relatively close meaning that there are consistent strong returns. Adventure, Fantasy, and Family also rank high on mean profit, but each shows a large mean-median gap, with Science Fiction being the worst aggressor of this with a mean of \$125.6M and a median of \$35.5M. Indicating their profitability is driven by a handful of blockbusters rather than average overall performance. Drama and Comedy are made far more often than any other genre with 1,441 and 1,110 films respectively, but rank lower in profitability, showing that volume doesn't mean good returns. Action combines both scale and profit with 918 films and a \$114.4M mean, while Western, Documentary, and especially Foreign are too small a sample to draw a real conclusions from. Taking all of this into account, Animation and Action stand out as the genres with the most potential, with Animation having consistent per-film profitability, and Action for having a good middle ground of profitability with production volume. While Adventure, Fantasy, and Science Fiction, have high averages being driven more by outliers than consistent profits.

4.2 Part II: Popularity vs. Audience Rating

Figure 5: Movie Popularity vs. Audience Rating
Color-coded by Net Profit

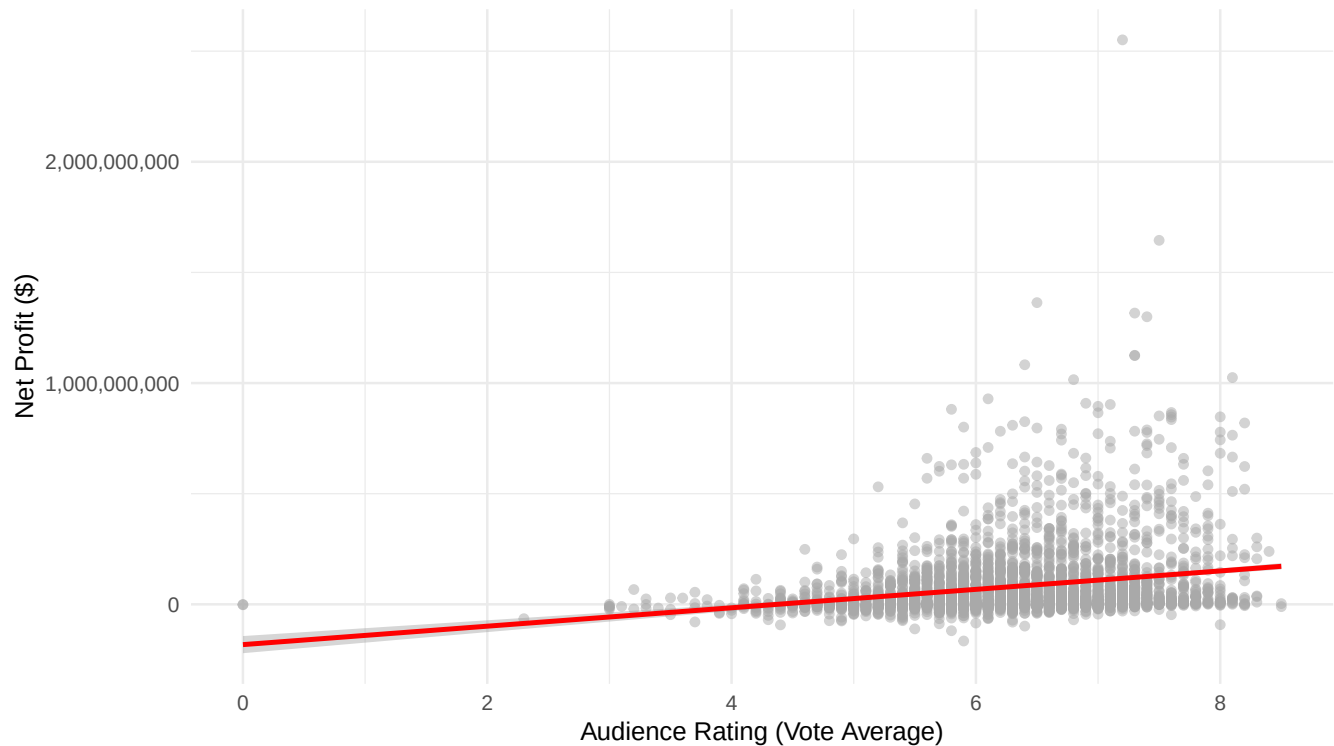


Analysis of Popularity vs. Rating:

The scatter plot and correlation matrix reveal that popularity and audience rating (vote_average) show only a weak positive relationship ($r \approx 0.29$). This means that widely watched and popular movies are not necessarily the best-received ones. Therefore, popularity often reflects the scope of marketing or the recognition of a franchise, rather than purely recognition from people or the completion of a movie. In addition, by coloring the dots differently according to their net profit, we were able to find a pattern. The most financially successful films, or brightly colored dots on the plot, are clustered in a comfortable rating range of 6.0 to 7.5. This means typical mainstream blockbuster films that prioritize popularity over critical acclaim. It should also be noted that movies with overrated ratings of 7.5 are not short on profit. In fact, these masterpieces tend to outperform the average of the entire dataset in both popularity and revenue. The point where the phenomenon that goes beyond our expectations is rather in movies low in ratings. This is because there are cases where some films with poor reviews still earn high popularity scores and earn profits only due to external factors such as great expectations or brand recognition (IP) before release.

4.3 Part III: Audience Rating vs. Net Profit

Figure 6: Net Profit vs. Audience Rating
With Ordinary Least Squares (OLS) Trend Line



Pearson's product-moment correlation

```
data: movies_clean$vote_average and movies_clean$profit
t = 13.428, df = 3227, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.1971115 0.2624529
sample estimates:
      cor
0.2300414
```

Table 4: Pearson Correlation Test: Rating vs. Profit

Variable X	Variable Y	Correlation (r)	p-value
Vote Average	Net Profit	0.23	0

Analysis of Rating vs. Profit:

To test whether the true correlation between audience rating and profit is zero we must use a Pearson Correlation. Once we used the Pearson correlation we see that the coefficient was about $r = 0.23$, and the p-value was less than 0.0001 with a test-statistic of $t = 13.428$. Since the p-value is extremely small, we

reject the null hypothesis and conclude that the correlation is statistically significant, but its strength is not very large. Audience evaluation explains only about 5.3% of the variation in film revenue (r^2). A scatter plot also shows that many films with similar ratings earned very different revenues, indicating that audience ratings alone cannot accurately predict revenue. Some blockbuster films like Avatar and Star Wars earned very high box office revenues despite having ratings clustered around 7.0 to 8.0, and these films affect the fitted OLS trend line. Overall, higher audience ratings are associated with higher revenue, but that relationship is fairly weak.

Fitted line of best fit:

$$\hat{y} = -182,073,403 + 41,630,564 * x$$

where $\hat{y}$ = predicted net profit (\$), and x = audience rating (vote_average)

4.4 Conclusion

KEY FINDINGS SUMMARY

To conclude, the overall financial success is driven more by a film's marketing and visibility than by how audiences like it. With vote count at $r = 0.74$, popularity at $r = 0.59$, and budget at $r = 0.55$ we can see that these factors correlate the strongest with net profit, while audience rating at $r = 0.23$ rarely matters compared to the rest. However when looking at the question in the scope of ROI (profit / budget) we see that none of these factors provide a meaningful prediction, i.e. higher budgets generate bigger profits but no proportionally better returns on investment. When determining which genre aided the most in financial success we saw that Animation was the most consistent in profitability while Action was the best balance in sheer numbers and returns. Adventure, Fantasy and Science Fiction owe their high averages to a few blockbuster outliers rather than consistent overall performance. In tandem, these findings suggest for a small studio with limited resources prioritizing visibility and audience reach is more important than audience ratings, while also choosing genres with realistic returns for their investment.

LIMITATIONS

First of all what we have proven is correlation, not causation. These relationships don't establish that higher budget or popularity causes higher profit. Also factors like franchise recognition, release timing, marketing spending, and director/lead actors aren't captured in this dataset. A second limitation would be the zero/missing budget and revenue values. Where 1,574 movies, around 33%, were dropped for either zero or missing budget or revenue. This primarily affects older, low-budget films, and niche genres. There is also the issue of multi-genre movie, or a movie tagged with several genres which contributes to each genre's summary statistics, so genre comparisons aren't statistically independent of one another. There's also no inflation adjustment so budgets and revenues are taken without adjusting for inflation, which could inflate the profitability of more recent releases compared to older ones. Finally the dataset only includes movies notable enough to be catalogued by TMDb, so conclusions should not be generalized to all films ever produced.

References

- Used AI on JSON parsing and some Kable Functions. Also used to streamline code and help display functions neatly.
 - https://gemini.google.com/u/1/app?utm_source=app_launcher&utm_medium=owned&utm_campaign=base_all

- <https://chat.deepseek.com>
- Data Set Website:
 - https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata?select=tmdb_5000_movies.csv

Appendix

A.1 Setup & Package Loading

```
1 knitr::opts_chunk$set(message = FALSE)
2 knitr::opts_chunk$set(warning = FALSE)
3 knitr::opts_chunk$set(error = FALSE)
4 knitr::opts_chunk$set(echo = FALSE)
5 knitr::opts_chunk$set(fig.align = 'center')
6 library(tidyverse)
7 library(jsonlite)
8 library(janitor)
9 library(knitr)
10 library(kableExtra)
11 library(corrplot)
12 library(scales)
```

A.2 Loading and Cleaning the Data

```
1 movies_raw <- read_csv("C:/Users/Jiho/Desktop/PSTAT100/data/tmdb_5000_movies.csv") |> clean_names
2 colnames(movies_raw)
3 movies_clean <- movies_raw %>%
4   filter(budget > 0 & revenue > 0) %>%
5   mutate(
6     profit = revenue - budget,
7     roi = profit / budget,
8     release_year = year(as.Date(release_date))
9   )
```

A.3 Exploratory Data Analysis

```
1 movies_by_genre <- movies_clean %>%
2   mutate(genre_list = map(genres, ~ {
3     if (is.na(.x) || .x == "" || .x == "[]")
4       return(character(0))
5     res <- tryCatch(fromJSON(.x),
6                     error = function(e) NULL)
7     if (is.data.frame(res) &&
8         "name" %in% colnames(res)) res$name else character(0)
9   })) %>%
10  unnest(genre_list) %>%
11  rename(genre = genre_list)
```

A.4 Methodology: Correlation Matrix Setup

```
1 cat("Profit = Revenue - Budget")
2 cat("ROI = Profit / Budget")
3 cat("r = sum((x_i - x_bar) * (y_i - y_bar)) / sqrt(sum((x_i - x_bar)^2) * sum((y_i - y_bar)^2))")
```

```

4 cat("y_hat = B0 + B1 * x")
5 numeric_vars <- movies_clean |>
6   select(budget, popularity, vote_average, vote_count, profit, roi)
7
8 cor_matrix <- cor(numeric_vars, use = "complete.obs", method = "pearson")

```

A.5 Results, Part I: Factors vs. Profit

```

1 corplot(cor_matrix, method = "color", type = "upper", addCoef.col = "black",
2         tl.srt = 45,
3         title = "Figure 3: Correlation Between Movie Factors and Outcomes", mar = c(0, 0, 2, 0))
4 cor_matrix |>
5   as.data.frame() |>
6   rownames_to_column("Variable") |>
7   kable(digits = 2, caption = "Pairwise Correlations Between Movie Factors and Outcomes")
8 genre_summary <- movies_by_genre[!is.na(movies_by_genre$genre), ]
9
10 genre_list <- unique(genre_summary$genre)
11
12 results <- data.frame(genre = character(), n_movies = numeric(),
13                      mean_profit = numeric(), median_profit = numeric(),
14                      mean_popularity = numeric())
15
16 for (g in genre_list) {
17   temp <- genre_summary[genre_summary$genre == g, ]
18   results <- rbind(results, data.frame(
19     genre = g,
20     n_movies = nrow(temp),
21     mean_profit = mean(temp$profit, na.rm = TRUE),
22     median_profit = median(temp$profit, na.rm = TRUE),
23     mean_popularity = mean(temp$popularity, na.rm = TRUE)
24   ))
25 }
26
27 genre_summary <- results[order(-results$mean_profit), ]
28 genre_summary$profit_millions <- genre_summary$mean_profit / 1000000
29
30 ggplot(genre_summary, aes(x = genre, y = profit_millions, fill = mean_profit)) +
31   geom_col() +
32   coord_flip() +
33   theme_minimal() +
34   scale_fill_distiller(palette = "Oranges", direction = 1) +
35   labs(title = "Figure 4: Average Net Profit by Genre", x = "Genre", y = "Mean Net Profit (Million
36 genre_summary |>
37   select(genre, n_movies, mean_profit, median_profit, mean_popularity) |>
38   kable(digits = 0, caption = "Genre-Level Summary: Movie Count, Profit, and Popularity",
39         col.names = c("Genre", "N Movies", "Mean Profit", "Median Profit", "Mean Popularity"))

```

A.6 Results, Part II: Popularity vs. Rating

```

1 movies_clean %>%
2   ggplot(aes(x = vote_average, y = popularity,
3             color = profit)) +
4   geom_point(alpha = 0.7, size = 1.5) +
5   scale_color_viridis_c(option = "plasma",
6                        labels = comma, name = "Net Profit ($)") +
7   labs(
8     title = "Figure 5: Movie Popularity vs. Audience Rating",
9     subtitle = "Color-coded by Net Profit",
10    x = "Audience Rating (Vote Average)",
11    y = "Popularity Score"
12  ) +
13  theme_minimal() +
14  theme(legend.position = "right")

```

A.7 Results, Part III: Rating vs. Profit

```

1 movies_clean %>%
2   ggplot(aes(x = vote_average, y = profit)) +
3   geom_point(alpha = 0.5, color = "darkgrey") +
4   geom_smooth(method = "lm", color = "red", se = TRUE) +
5   scale_y_continuous(labels = comma) +
6   labs(
7     title = "Figure 6: Net Profit vs. Audience Rating",
8     subtitle = "With Ordinary Least Squares (OLS) Trend Line",
9     x = "Audience Rating (Vote Average)",
10    y = "Net Profit ($)"
11  ) +
12  theme_minimal()
13 cor_test_result <- cor.test(movies_clean$vote_average,
14                           movies_clean$profit, method = "pearson")
15 cor_test_result
16 tibble(
17   `Variable X` = "Vote Average",
18   `Variable Y` = "Net Profit",
19   `Correlation (r)` = cor_test_result$estimate,
20   `p-value` = cor_test_result$p.value
21 ) %>%
22   kable(digits = 4, caption = "Pearson Correlation Test: Rating vs. Profit") %>%
23   kable_styling(latex_options = "hold_position")
24 rating_profit_lm <- lm(profit ~ vote_average, data = movies_clean)
25 lm_coefs <- coef(rating_profit_lm)
26
27 cat(sprintf("Fitted line of best fit:\ny_hat = %s + %s * x\n\nwhere y_hat = predicted net profit (")
28       comma(round(lm_coefs[1], 2)),
29       comma(round(lm_coefs[2], 2))))

```